

Article

Synthesis and Properties of Self-Polishing Antifouling Coatings Based on BIT-Acrylate Resins

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Abstract: Painting antifouling coatings is one of the most important methods to prevent marine biofouling. Acrylic resin is widely used in marine antifouling because of its excellent stickiness, water resistance, and film-forming capabilities. At present, the widely used acrylate antifouling coatings require a high concentration of cuprous oxide as antifoulant. The release and accumulation of copper ions are the main factors affecting the marine environment. In this study, BIT-allyl methacrylate (BM) and zinc acrylate (ZM) were selected as functional monomers copolymerized with methyl methacrylate (MMA) and butyl acrylate (BA) to prepare a series of BIT acrylate antifouling resins. The inhibitory effects of all resins against marine bacteria (*S. aureus*, *V. coralliilyticus*, and *V. parahaemolyticus*), marine algae (*Chlorella*, *I. galbana*, and *C. curviusetus*), and barnacle larvae were studied. Moreover, marine field tests on the BIT modified resin in coastal waters were conducted. The results demonstrate that the grafted BIT-zinc acrylate resin not only exhibits excellent antifouling properties but also a significant self-polishing performance, providing a novel strategy to design a long-term antifouling resin with stable antifoulant release.

Keywords: self-polishing; zinc acrylate resins; antibacterial; anti-algal; antifouling coating



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1. Introduction

Numerous fouling organisms living in the ocean adhere and grow on the surface of solid objects, such as boats and ships, submarines, aquaculture facilities, underwater drilling facilities, etc. Marine biofouling is prompted by the adhesion and reproduction of organisms [1,2]. The fouling by marine organisms destroys the surface of facilities and underwater equipment, reduces the stability and safety of marine equipment, and imposes economic losses [3–5]. Antifouling coatings, particularly self-polishing antifouling coatings, are currently the most cost-efficient and effective strategies to inhibit the adhesion of marine organisms [6,7]. The traditionally used organotin self-polishing antifouling coating offers long-term stability, stable hydrolysis, and a good antifouling effect; however, it is highly toxic to fish and other marine organisms and leads to a substantial increase in pollution to the marine environment. Organotin coatings, on the other hand, cannot dissolve naturally and are enriched in saltwater, or they might enter the human body through the food chain, causing serious health problems [8,9]. Therefore, the development of novel low-toxic/non-toxic environmentally friendly antifouling coatings has emerged as an important research area in the field of marine antifouling coatings.

Currently, a variety of novel marine antifouling coatings are being developed, including low-surface-energy coatings [10–12], antifoulant grafted coatings [13,14], amphiphilic coatings [15,16], main chain degradable coatings [17,18], the resistance of protein adsorption coatings [19,20], biomimetic surface microstructure coatings [21,22], nanocomposite

coatings [23,24], hydrogel coatings [25–27], and other dynamic surface coatings, etc. However, as a typical representative of self-polishing technology, the widely used tin-free self-polishing coating remains the primary antifouling approach with outstanding antifouling efficacy, a longer-term validity period, and more cost-effectiveness.

In recent years, various tin-free self-polishing antifouling coatings, such as acrylic copper [28], acrylic zinc [29], and acrylic silicone [30], have been explored and applied in the marine antifouling areas. Tian et al., developed a micron-level Cu–Ti composite antifouling coating with high mechanical durability and the efficiency against bacteria almost reached 100% [28]. Despite its great antifouling performance, copper antifouling coatings continuously release and accumulate copper ions in docks and ports, causing a hazard to the marine environment and species. Therefore, zinc acrylate resin as a self-polishing antifouling coating has attracted much attention from researchers due to its low toxicity and environmental friendliness. Kim et al., synthesized a series of zinc acrylic copolymers (Zn–SPC) and investigated their polishing properties [29]. Chen et al., studied a zinc–silicon acrylic copolymer and observed that the resin exhibited outstanding inhibitory effects on *Phaeodactylum tricornutum* (*P. tricornutum*) [30]. Further research demonstrated that the copolymer could be used as a marine antifouling under static or low-flow conditions. Dai et al., described a type of zinc–polyurethane copolymer, for which the maritime field test demonstrated an outstanding self-polishing antifouling effect within 12 months, while the biocides were constantly discharged [31].

In this study, we selected 1,2-benzisothiazol-3(2H)-one (BIT) as an effective antifouling ingredient, and modified it to obtain a polymerizable BIT–allyl methacrylate (BM). A series of acrylic zinc copolymers were synthesized by copolymerization of methyl methacrylate (MMA), butyl acrylate (BA), BM, and acryloyloxy methacryloxy zinc monomers (ZM) (Scheme 1). To improve adhesion to the substrate, hard monomer MMA and soft monomer BA monomer were utilized to modify the mechanical characteristics of the polymer. Marine fouling organisms were inhibited by the ZM monomer used to enhance the polishing quality and to ensure the slow release of antifoulant contained in the coating. The structures of the copolymers and their inhibitory effects on marine bacteria and marine algae were investigated, along with the antifouling performance by the inhibitions of barnacle larvae and marine field test when used as an antifouling coating.

2. Materials and Methods

2.1. Synthesis of Monomers and Polymers

2.1.1. Synthesis of BIT Monomer (BM)

Triphosgene (9.9 g, 0.033 mol) and toluene (20 mL) were mixed in a 100 mL reaction flask and vigorously stirred for 10 min at room temperature, followed by cooling to 5 °C and a mixture of allyl alcohol (6.22 g, 0.107 mol), toluene (25 mL), and trimethylamine (TEA) (0.5 mL) was slowly added to the flask and the temperature was maintained at 5 °C. After addition, the mixture was stirred for 0.5 h, before adding BIT (8.6 g, 0.057 mol) and heated to reflux for 2 h. The solution was transferred to a separate funnel and the pH was adjusted to neutral when the reaction was completed. Deionized water was used to wash the organic layer, which was then dried over anhydrous magnesium sulfate. Then, the organic layer was filtered and concentrated to obtain a white solid. The solid was subsequently recrystallized in cyclohexane and the yield of BM monomer was 80%. The melting point of the BM monomer is 113–115 °C. ¹H NMR (400 MHz, CDCl₃), 7.94 (d, H, PhH), 7.62 (t, H, PhH), 7.46 (d, H, PhH), 7.33 (t, H, PhH), 5.97 (m, H, CH), 5.46 (d, H, CH₂), 5.29 (d, H, CH₂), 4.83 (d, 2H, CH₂). IR (KBr), ν (cm^{−1}): 3091, 2948 (C–H), 1725 (C=O), 1650, 1590 (C=C), 1452 (CH₂), 1280, 1099 (C–O–C), 981, 921 (C–H), 740(C–H).

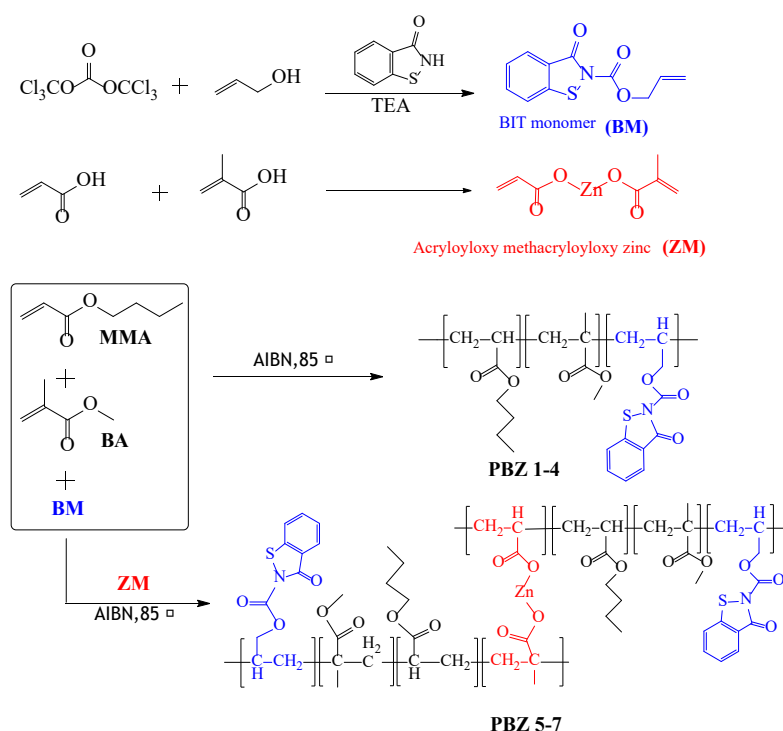
2.1.2. Synthesis of Acryloyloxy Zinc Monomer (ZM)

Firstly, 5.00 g (0.05 mol) of freshly manufactured zinc hydroxide and 35 mL of xylene were mixed into a 100 mL three-necked flask, ultrasonically dispersed for 30 min, and then heated to 70 °C. An acrylic acid (3.60 g, 0.05 mol), methacrylic acid (5.17 g, 0.06 mol), and

xylene 15 mL mixture, mixed in a constant pressure funnel, was added drop by drop to the hydroxide solution. After the dropwise addition was completed, the mixture was stirred at room temperature for one hour, and then filtrated under reduced pressure to obtain a white powder. The crude product was purified by dissolving it in CH_2Cl_2 to remove insoluble impurities. IR (KBr), ν (cm^{-1}): 3099, 2972 (C–H), 1643 (C=C), 1600 (O–Zn–O), 1445, 1373 (CH_3), 1242, 1070 (C–O–C), 979, 943 (C–H).

2.1.3. Synthesis of Copolymers

The xylene and *n*-butanol mixture was heated to 85 °C in a 100 mL flask equipped with a condenser and a thermometer. The monomers of BM, MMA, BA, and ZM were dissolved in xylene and *n*-butanol according to their stoichiometric ratios, and after adding 2,2'-azobis(2-methylpropionitrile) (AIBN) (4.5–5.0 wt % of the total monomers) and dissolving it completely, the mixture was transferred to the above flask in 3 h and stirred continuously for another 3 h at 85 °C. TLC chromatography was utilized to monitor the reaction process and to determine whether the BM monomer disappeared. After the reaction was completed, the mixture was cooled and a golden yellow transparent polymeric solution was obtained. The synthetic route of the copolymers is displayed in Scheme 1.



Scheme 1. Synthesis of self-polishing zinc acrylate copolymers.

2.2. Structural Characterization

The structures of monomers and the polymers were analyzed using an NMR spectrometer (BRUKER av-400) and Fourier transform infrared spectrometer (TENSOR-27). The molecular weight was determined by gel permeation chromatography (Waters 1515, Milford, MA, USA). A certain amount of polymer was dissolved in 1 mL of tetrahydrofuran to make a concentration of approximately 5 mg/mL for determination. Tetrahydrofuran was used as the mobile phase, the column temperature was 35 °C, and the flow rate was 1 mL/min. Calibration was performed using polystyrene as a standard. The glass transition temperature T_g of the resulting polymer was determined by DSC (Q100 SDT, TA Inc., New Castle, DE, USA). AQ600 SDT (TA Inc., New Castle, DE, USA) thermogravimetric analyzer was utilized to test the thermal decomposition of polymers and the contact angle was

measured using an Optical Contact Angel and Interface Tension Meter (SL200KB, Shanghai SUOLUN TECH, Shanghai, China).

2.3. Weight Loss Measurement

The self-polishing performance of the polymers was determined by weight loss in the natural seawater at room temperature. After painting the polymers to the surface of the glass plate and allowing them to dry, the slides were fitted on a laboratory dynamic water washout simulator with the rotation speed of 3 m/s. The plates were weighed once a week, after continuous scouring. The formula for calculating the mass loss rate is as follows:

$$\text{Mass loss (wt\%)} = \frac{M_0 - M_t}{M_0 - M_b} \times 100\% \quad (1)$$

where M_0 is the mass of the initial sample plate, M_t is the mass of the plate after the scouring time interval t , and M_b is the mass of the blank plate (Blank).

2.4. Algae Inhibition Activity Test

The algae *Chlorella*, *I. galbana*, and *C. curvisetus* were cultivated and supplied by Ocean College of Hainan University. Resins were coated on the glass plate (76.2 mm × 25.4 mm × 1.2 mm ± 0.5) and dried at room temperature. We needed to sterilize the sample plate under UV light for 2 h before testing to avoid the influence of other impurities. The plate was then placed in a glass cup and immersed in diluted algal solution with an absorbance between 0.06 and 0.1 Abs. At room temperature, the cup was sealed with a semi-permeable membrane and illuminated by daylight lamp with a light intensity of 5000–6000 lx. Every 24 h, the absorbance of algae solution was measured and compared with the blank glass plate. Using the recorded absorbance data, with abscissa as time (d) and ordinate as algal concentration (C), a graph was obtained using the linear regression equation of absorbance value–algal solution concentration. The seven-day growth inhibition rate of each algal by resins was [32] and the formula for calculating the inhibition rate is as follows:

$$S = \frac{C_0 - C_i}{C_0} \times 100\% \quad (2)$$

where, S is the growth inhibition rate of algae, C_0 is the concentration of the initial algae solution, and C_i is the concentration of the algae-solution-immersed polymer.

2.5. Antibacterial Activity Test

S. aureus, *V. coralliilyticus*, and *V. parahaemolyticus* were typical marine bacteria provided by the Marine Drug Research Laboratory of Hainan University. *S. aureus* was cultured with tryptone soy broth (TSB), while *V. coralliilyticus* and *V. parahaemolyticus* were cultured with 2216E liquid medium. The antibacterial effect of the resin was studied via the absorbance method. We needed to sterilize the sample plate under UV light for 2 h before testing to avoid the influence of other impurities. The resin was coated and evenly dispersed on the cell culture plate (6 wells), dried at room temperature, and then the culture solution and bacterial solution were added to the coated culture plate, where bacteria were cultured at a constant temperature of 37 °C. The control group is the hole with only culture medium and bacteria added. All the water used in the antibacterial test was ultrapure water, not seawater. The bacterial solution was sucked into a 96-well cell culture plate and the absorbance at 600 nm was measured for 2 h, 4 h, 6 h, 8 h, 10 h, and 12 h in a microplate reader, and the absorbance changes in the bacterial solution were recorded.

Taking abscissa as time (d) and ordinate as absorbance (A), the absorbance–time graph was obtained. Calculating the 12 h growth inhibition rate of each sample to bacterial solution, the formula for calculating the inhibition rate is as follows:

$$A = \frac{A_0 - A_i}{A_0 - 0.047} \times 100\% \quad (3)$$

where, A is the growth inhibition rate of bacteria, A_0 is the maximum concentration of the blank bacterial solution, and A_i is the concentration of the sample bacterial solution.

2.6. Inhibition of Barnacle Larval Activity Test

A large number of adult barnacles needed to be collected, cleaned, and dried in a dark place for 12 h. Then, barnacles were placed in seawater under light conditions and a large number of larvae were released for further experiments [33]. We needed to sterilize the sample plate under UV light for 2 h before testing to avoid the influence of other impurities. The resin was painted evenly in a 90 mm cell culture dish and dried at room temperature, after which, 20 active barnacle larvae were placed in the dish and the activity of barnacle larvae was observed and recorded after 24 h, and then the inhibition rate was calculated using the following formula:

$$N = \frac{N_0 - N_t}{N_0} \times 100\% \quad (4)$$

where, N represents the growth inhibition rate of barnacle larva, N_0 is the initial number of barnacle larvae, and N_t , the number of attached barnacle larvae at 24 h.

2.7. Marine Field Test

The marine antifouling properties of the polymeric coating were evaluated in Haikou Bay (20°05' N, 110°15' E) of Hainan Province from May to August. The test site is home to many marine fouling organisms, mainly bacteria, algae, ascidians, lime worms, oysters, mussels, and barnacles. The water temperature is between 30–34 °C and the salinity is between 29.6–31.8‰. According to the national standard GB 5370-2007, the polymer coated steel plates (300 × 200 × 3 mm³) were fixed to a stainless steel frame and were placed 1–1.5 m below the water surface. After immersion for a time interval, the plates were periodically removed from the seawater, the slit surface was washed with seawater gently, photographs were taken, and the antifouling activity was evaluated according to the standard ASTM D6990-05 (2011).

3. Results and Discussion

3.1. Characterization

The copolymers were synthesized by incorporating the BIT monomer, acrylate monomer, and zinc acrylate monomer in different proportions (Table 1). The number average molecular weight (M_n) of the resins ranged from 9800–14,100, while the weight average molecular weight (M_w) was 10,800–14,300. The difference in molecular weight affected the leaching rate of the self-polishing coating. The polymers' dispersion index (PDI) ranged between 1.01 and 1.14, implying a relatively uniform molecular weight distribution. Each copolymer exhibits a single glass transition temperature (T_g), indicating that all resins were random copolymers.

Table 1. Molecular compositions and characterizations of copolymers.

Samples	Copolymer Composition (%) BM/MMA/BA/ZM	M_n (g/mol)	M_w (g/mol)	PDI	T_g (°C)
PBZ0	0/50/50/0/0	9858	11,280	1.14	58.03
PBZ1	5/47.5/47.5/0/0	10,979	11,334	1.03	72.37
PBZ2	10/45/45/0/0	10,294	11,839	1.15	67.52
PBZ3	15/42.5/42.5/0/0	11,516	12,662	1.1	65.69
PBZ4	20/40/40/0/0	11,317	12,701	1.12	56.2
PBZ5	20/40/40/0/10	11,184	11,380	1.02	65.34
PBZ6	20/40/40/0/15	11,856	11,965	1.01	68.68
PBZ7	20/40/40/0/20	14,099	14,298	1.01	64.55

The structure of the copolymers (for example polymer PBZ4) was confirmed by FT-IR spectrometer (Figure 1a). The absorption peak at 3087 cm^{−1}, attributed to Csp²-H stretching

vibration of the monomer BM, disappeared in PBZ4, and the characteristic absorption peaks at 1650, 1590, and 1452 cm^{-1} , attributed to the aromatic ring existing in the BM, appeared in PBZ4, indicating that the BM monomer had completely polymerized with MMA and BA. Meanwhile, the characteristic FT-IR peaks at 981 and 921 cm^{-1} , attributed to the deformation vibration of Csp²-H in BM, disappeared in PBZ4, and the characteristic peak at 740 cm^{-1} , a result of the aromatic ring Csp²-H bending vibration in BM, also appeared in PBZ4, demonstrating that the monomer BM was integrated into the polymer.

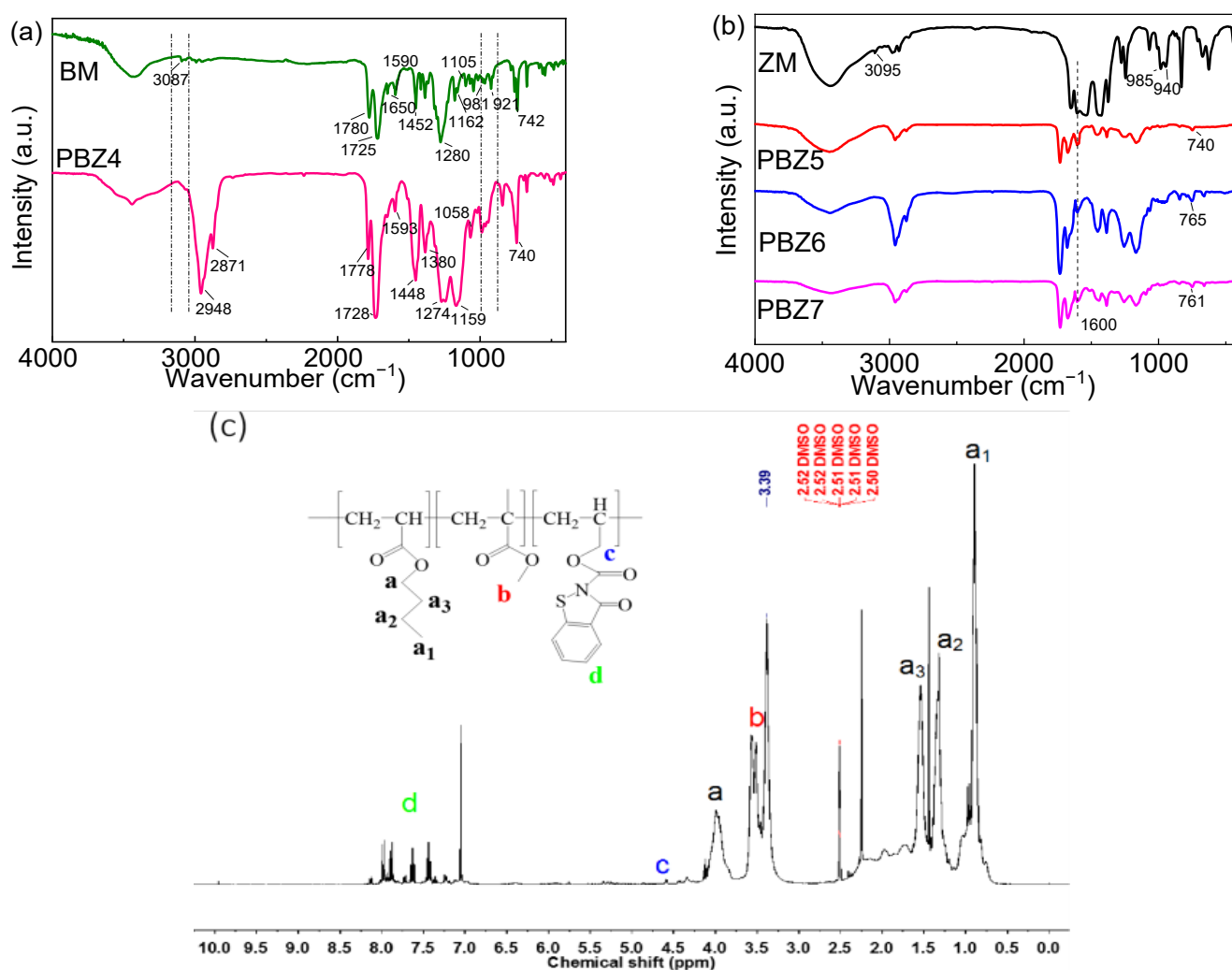


Figure 1. (a) FT-IR spectrum of BM monomer and polymer PBZ4; (b) FT-IR spectrum of ZM monomer and polymer PBZ5-7; (c) ¹H NMR spectrum of polymer PBZ4.

Similarly, the FT-IR spectra of polymers PBZ5-7 compared to ZM are displayed in Figure 1b. The disappearance of the absorption peak at 3095 cm^{-1} due to the Csp²-H stretching vibration of the ZM monomer disappeared in the polymers, as along with the disappearance of the absorption peaks at 985 and 940 cm^{-1} due to the Csp²-H bending vibration of the ZM monomer in polymers, indicated that the ZM monomer had polymerized with MMA and BA. Meanwhile, the absorption peaks at 3100–3000 cm^{-1} and 990–910 cm^{-1} that appeared in the ZM monomer disappeared in PBZ polymers, indicating that the ZM monomer had completely copolymerized. Furthermore, the presence of absorption peaks near 750 cm^{-1} attributed to the Csp²-H bending vibration of benzene ring in PBZ5-7 indicated that the copolymers contain monomer BM fragments.

The chemical structure of polymer PBZ4 was analyzed by ¹H NMR (Figure 1c). There were four groups of characteristic peaks at δ = 8.25–7.25, which belonged to the characteristic

peaks of hydrogen in the benzene ring. This showed that the copolymer contains a small amount of the benzene ring, which corresponds to the d part in the BM monomer. A group of characteristic peaks appeared at $\delta = 4.75\text{--}4.5$, which belonged to the characteristic peaks of hydrogen in methyleneoxy. It shows that the copolymer contains a methyleneoxy group, which corresponds to the c part in the BM monomer. The CH in the acrylate $-\text{COOCH}_2\text{R}$ is in the deshielding region of the ether oxygen atom, and the position of the peak shifts to the lower field. Two sets of characteristic peaks appeared at $\delta = 4.25\text{--}3.75$ and $\delta = 3.60\text{--}3.40$, which belonged to the characteristic peaks of CH in RCOOCH_3 and RCOOCH_2R . This showed that the copolymer contains methyl acrylate and butyl acrylate, corresponding to part a in methyl acrylate and part b in butyl acrylate. Several groups of characteristic peaks appeared at $\delta = 1.75\text{--}0.5$, which belonged to the characteristic peaks of H in CH_3 and CH_2 under general environment. This demonstrated that the resin contains methyl and methylene, corresponding to the a1, a2, and a3 parts of butyl acrylate. The above analysis confirmed that the polymer PBZ4 was synthesized.

Figure 2a depicts the thermal decomposition curves of copolymers. The decomposition temperature of all the copolymers was higher than 200°C , making them appropriate for the variable marine environment. However, the thermal stabilities of the polymer PBZ7, grafted with 20% monomer ZM, decreased noticeably with the decomposition temperature less than 150°C . As the temperature increased, the weight loss rate was approximately 30% in $150\text{--}250^\circ\text{C}$, and the rate was approximately 50% in $250\text{--}500^\circ\text{C}$. Given that metal zinc cannot be vaporized, the weight loss rate of polymer PBZ7 can only reach 80%. The weight loss rate of polymers PBZ0 and PBZ4 was almost 100%. The TGA curves were consistent with typical characteristics of thermal decomposition trends of acrylic polymers [34,35].

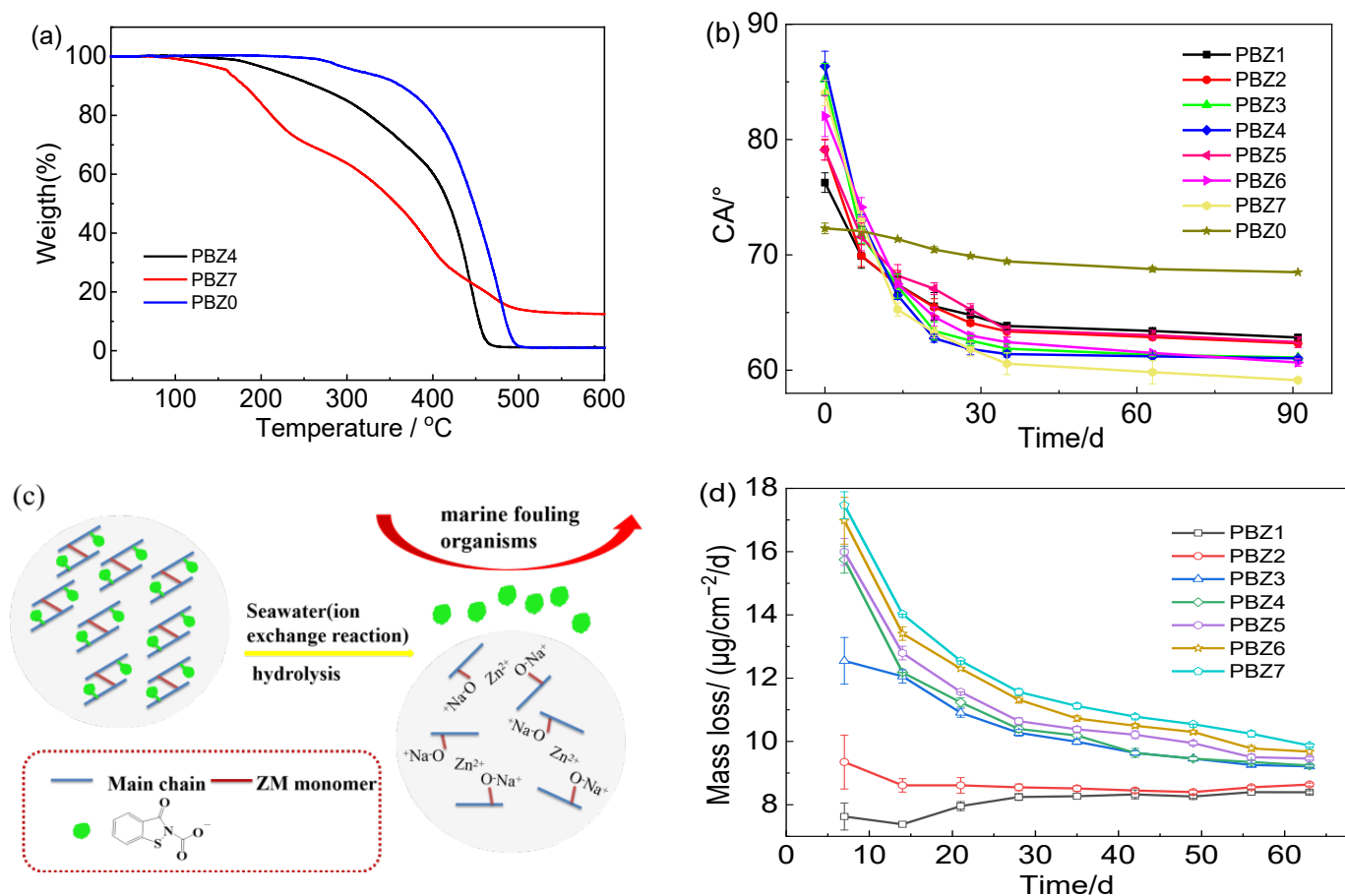


Figure 2. (a) Thermogravimetric curve. of copolymer PBZs, (b) contact angle change of copolymer PBZs, (c) the hydrolysis procedure of copolymer PBZs, (d) mass loss in hydrolysis of copolymer PBZs.

Figure 2b illustrates the change of the contact angle by immersing the resins in seawater. The initial contact angle of resins PBZ1-7 was greater than that of the control sample PBZ0, and the addition of monomer BM and ZM increased the hydrophobicity, while increasing the amount of monomer meant that contact angle values were increased. The contact angle of resins PBZ1-7 decreased remarkably after immersion in seawater. However, the value of PBZ0 decreased slowly. Due to the fact that the hydrolysis of BM and ZM produced hydroxyl and carboxyl groups, the polymer's surface was covered by a hydrophilic layer and the contact angle decreased, while the hydrophobicity increased. After 30 days, the hydrolysis rate of terminal group became stable and the value of contact angle became steady.

The hydrolysis properties of BIT–zinc acrylate polymers based on the cleavage of O–Zn–O linkage and the ester bond of BM monomer are shown in Figure 2c. In seawater, the antifoulant was in the form of BIT–formic acid and the zinc ion was replaced by sodium and potassium to generate hydrophilic groups. Antifoulant was continuously released as hydrolysis progressed to create antifouling effects. When the hydrophilic group reaches a certain concentration, it causes coating surface erosion and forms a newer flat layer of resin [36].

The weight loss of the resins is attributed to the self-polishing of zinc acrylate and the hydrolysis of BM fragments. After 7 days of seawater immersion, the weight of the polymers decreased significantly (Figure 2d). With the concentration of BM monomer increasing from 5% to 20%, the weight loss rate increased gradually, indicating that the hydrolysis of the polymers PBZ1-4 increased as the concentration of hydrolyzable monomer increased. In addition, it was observed that the weight loss of the polymers was proportional to their ZM monomer content—the higher the content of ZM, the greater the weight loss rate—demonstrating that the addition of ZM monomer improved the hydrolysis ability of the copolymers. For example, the average weight loss of polymer PBZ1 was $8.4 \mu\text{g}/\text{cm}^{-2}/\text{d}$, and the average weight loss of polymer PBZ7 rose to $9.9 \mu\text{g}/\text{cm}^{-2}/\text{d}$.

3.2. Algae Inhibition Activities

The working curve for algae absorbance–concentration refers to the previous experimental results [37]. According to the linear relationships between the absorbance algae concentration, the algae inhibition of PBZs (*Chlorella*, *I. galbana* and *C. curvisetus*) shown in Figure 3a–c, and the inhibitory rate of PBZs against the algae are displayed in Figure 3d.

PBZ1-7 inhibited algae growth exceptionally well, and the reproduction rate was significantly lower than the pure acrylic resin (PBZ0) and blank plate. The algae concentration increased rapidly in the control group PBZ0 and Blank aqueous solution, but remained almost unchanged in the PBZ1-7 solutions. The algal concentration decreased as the grafted BM monomer content increased—when the BM monomer content reached 20%, the growth and reproduction of algal was almost inhibited completely.

When the BM monomer content was kept at 20% and the ZM monomer content was increased from 5% to 15%, the inhibitory performance of polymers PBZ5-7 on algae increased continuously, which could be attributed to the hydrolysis and release of zinc ions from polymers, which promoted almost no inhibitory effect on the growth of the aforementioned algae. The inhibition rates of polymer PBZ7 on *Chlorella*, *I. galbana*, and *C. curvisetus* were 92.93%, 99.43%, and 99.58%, respectively. On the other hand, the pure acrylic polymer PBZ0 hardly had any inhibitory effects on the growth of the above algae. The hydrolysis and release of small molecular compounds from the side chains of the copolymers inhibit algal growth and the small molecular compound from side chains of the copolymers inhibits algal growth, and the small molecular compound enters cell interior by breaking the algal cell wall and combines with the bases on the cell nucleic acid, resulting in the cell rupture and inhibiting or completely killing the algae [38,39].

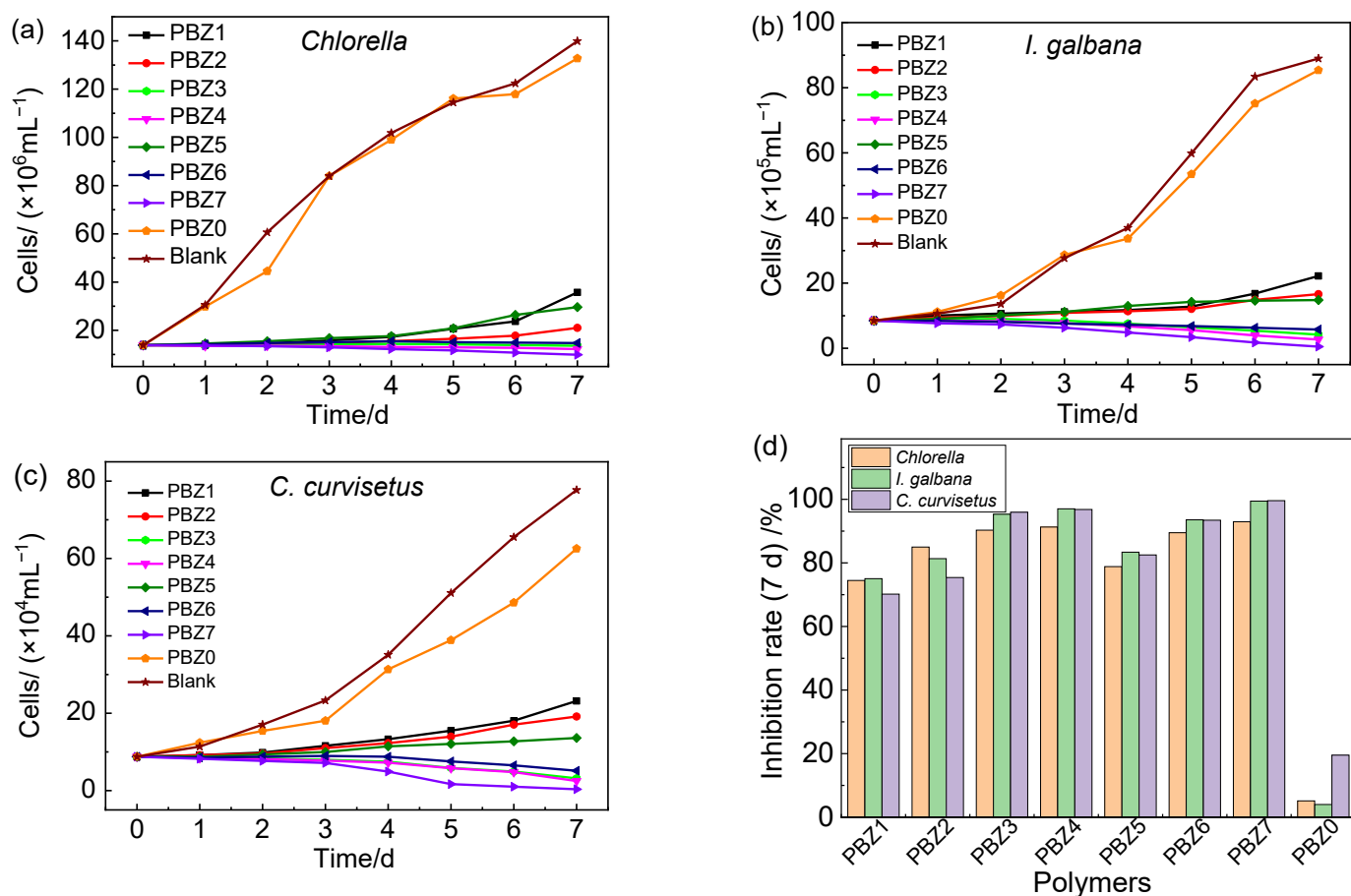


Figure 3. Inhibition of PBZs on *Chlorella* (a), *I. galbana* (b), and *C. curvisetus* (c). Inhibition rate of copolymer PBZs against the growth of algae (d).

3.3. Antibacterial Activities

The inhibitory properties of the copolymer PBZs on the growth of three marine bacteria (*S. aureus*, *V. coralliilyticus*, and *V. parahaemolyticus*) are shown in Figure 4.

Figure 4 depicts the bacteriostasis of copolymers PBZ1–7, with the inhibition rate closely related to the content of the BM monomer. The bacteriostatic action increased proportionally as the concentration increased from 5% to 20%. The inhibition rate of PBZ4 on *S. aureus*, *V. coralliilyticus*, and *V. parahaemolyticus* reached 97.43%, 95.09%, and 95.81%, respectively. However, the inhibition of PBZ0 in the absence of monomer BM was lower than the grafted resins and the absorbance increased considerably after 2 h. The inhibition of acrylic polymers enhanced continuously when the polymerized bifunctional monomer ZM was added to the polymeric compound. The antibacterial rates of polymers PBZ7 with 20% ZM against the above three marine fungi increased to 99.80%, 99.39%, and 99.03%, respectively, which almost suppressed the fungi reproduction completely. The absorbance declined slowly after 2 h.

The hydrolysis of the copolymer side chain produces and releases a small molecule of BIT unit, which inhibits bacterial growth. The S–N bonds of the BIT break and react with the protein receptor to establish the S–S bond, and the S–S bond interacts with bases on the biological protein to form hydrogen bonds and firmly attached to bacterial cells, destroying the structure of DNA and making it impossible to replicate [40,41]. The synergistic effect of ZM monomer may be attributed to the zinc ion hydrolysis and released from polymers.

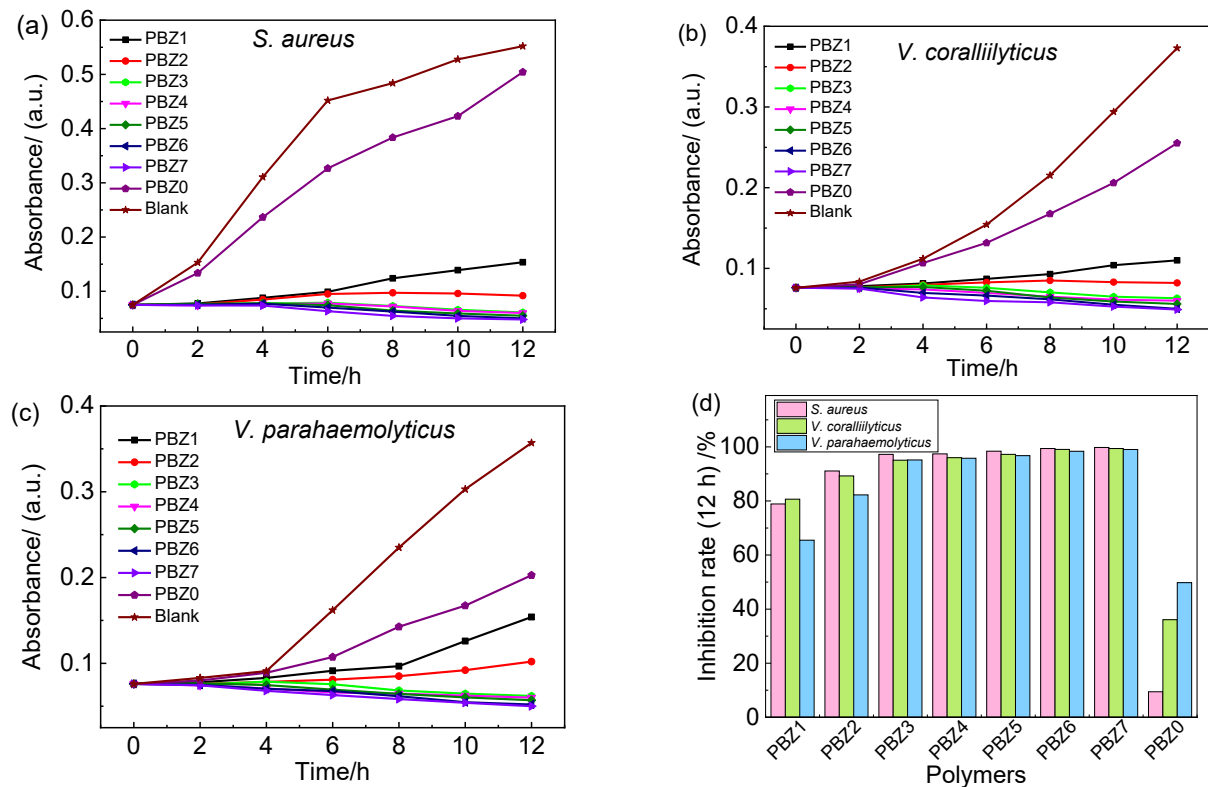


Figure 4. Antibacterial activity of PBZs against *S. aureus* (a), *V. coralliilyticus* (b), and *V. parahaemolyticus* (c). Inhibition rate of compounds against bacterial growth (d).

3.4. Inhibition of Barnacle Larval Activities

The inhibitory effects of resin PBZs on barnacle larvae are illustrated in Figure 5. Resins PBZ1-7 polymerized with BM monomers exhibited exceptional growth of barnacle larvae, with inhibition rates being substantially higher than that of pure polyacrylate PBZ0. When the component of BM monomer increased from 5% to 20%, the inhibition rates of polymers PBZ1-4 were 55.00%, 68.33%, 81.67%, and 88.33%, respectively. Furthermore, the polymers PBZ5-7 with 5% to 15% zinc acrylate monomer increased the rate of barnacle larvae to 78.33%, 95.00%, and 100.00%, respectively. Surprisingly, the inhibition rate of PBZ5 decreased compared to polymer PBZ4, and the inhibition rate of polymers PBZ6 and PBZ7 increased. The reason could be that the zinc element is an essential trace element for organisms, but it is also a toxic element and causes biological poisoning when its concentration exceeds the ecological range of organisms [42].

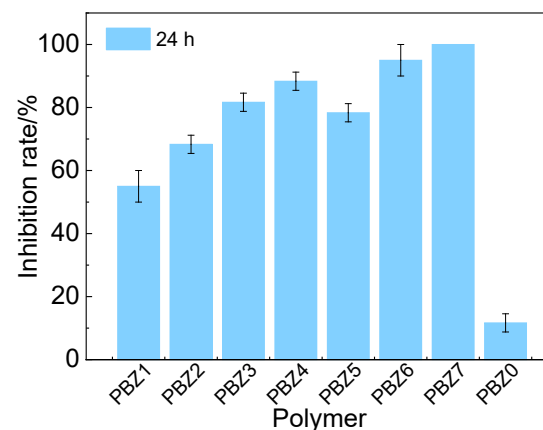


Figure 5. Inhibition rates of barnacle larva inhibited by PBZs.

3.5. Field Studies as Antifouling Coatings

To further study the inhibition potency of the BIT copolymers, the application performances of the copolymers as antifouling coatings were evaluated in marine field tests. Polymers were painted on the surface of steel plates and then immersed in natural seawater at boom season for 3 months (typically May–August), during which time the marine creatures adhering to the sheet steel were recorded periodically and polyacrylate (Blank) was selected as control group for contrast. We cropped the images of the tested panels after immersion in seawater, saved the $15 \times 20 \times 3 \text{ mm}^3$ in the middle of the plates as the typical, and the images are displayed in Figure 6.

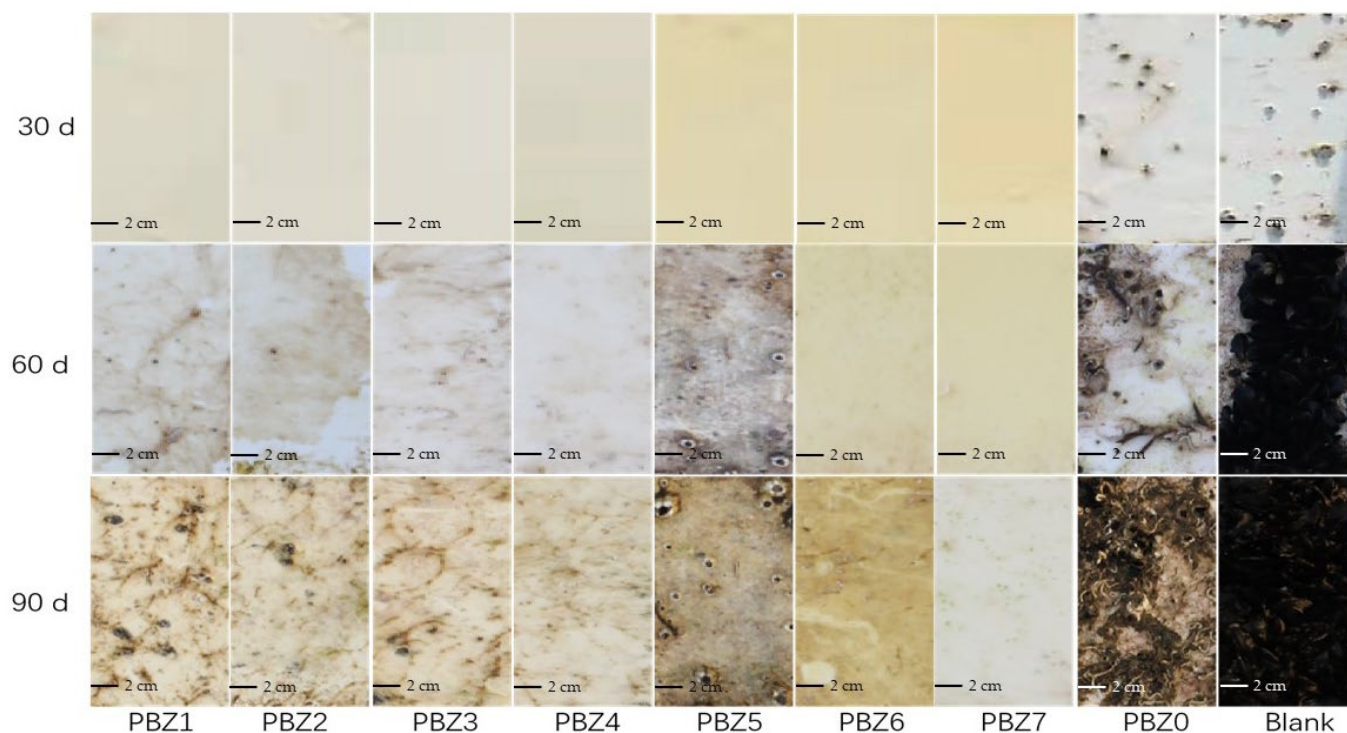


Figure 6. Antifouling performance of the resin immersion in seawater.

All of the evaluated resins outperformed the control group in terms of antibiofouling activity. After 30 days of immersion in saltwater, no evident fouling organism adhered to the polymer PBZs, with only a small number of barnacles colonizing on the surface of pure polyacrylate PBZ0 and the blank plate. After 60 days, a small number of algae began to adhere on the polymers, but few large-scale marine organisms, such as barnacles and mussels, adhered to the surface. However, the surface area of barnacles adhering to pure polyacrylate PBZ0 increased and a small amount of lime worms were observed. The blank plate was densely covered with barnacles and mussels, and some lime worms had grown in the gap between the mussels.

After 90 days of immersion in seawater, a few fouling organisms attached on the surface of the copolymeric resins PBZ1–7, the predominant organisms were algae and a few barnacles. As the BM monomer concentration increased from 5% to 20%, the antifouling activity improved proportionally. Moreover, the addition of ZM monomer also improved the antifouling performance of the coating. The blank plate and the pure polyacrylate PBZ0, on the other hand, were heavily adhered to by a large number of mussels and barnacles.

4. Conclusions

A series of BIT copolymerized polyacrylate resins PBZs were prepared, the structures were confirmed by ^1H NMR, FTIR, and GPC, while the inhibitory effects of the resins on marine algae, bacteria, and barnacle larvae were studied and discussed. The results

demonstrated that the BIT copolymers exhibited significant antifouling properties and the antifouling performance improved with increasing BM monomer or ZM monomer concentration. The 24 h inhibition rate was about 100% when the resin grafted 20% BM monomer and 15% ZM monomer. The field application studies also revealed that the co-polyacrylate resins possessed excellent antifouling properties. This work provided a theoretical foundation for further study of environmentally friendly and effective self-polishing antifouling coatings.

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References

1. Ciriminna, R.; Bright, F.V.; Pagliaro, M. Ecofriendly Antifouling Marine Coatings. *ACS Sustain. Chem. Eng.* **2015**, *3*, 559–565. [\[CrossRef\]](#)
2. Gu, Y.; Yu, L.; Mou, J.; Wu, D.; Xu, M.; Zhou, P.; Ren, Y. Research Strategies to Develop Environmentally Friendly Marine Antifouling Coatings. *Mar. Drugs* **2020**, *18*, 371. [\[CrossRef\]](#) [\[PubMed\]](#)
3. Callow, J.A.; Callow, M.E. Trends in the development of environmentally friendly fouling-resistant marine coatings. *Nat. Commun.* **2011**, *2*, 244. [\[CrossRef\]](#) [\[PubMed\]](#)
4. Liu, M.; Li, S.; Wang, H.; Jiang, R.; Zhou, X. Research progress of environmentally friendly marine antifouling coatings. *Polym. Chem.* **2021**, *12*, 3702–3720. [\[CrossRef\]](#)
5. Tian, L.; Yin, Y.; Bing, W.; Jin, E. Antifouling Technology Trends in Marine Environmental Protection. *J. Bionic Eng.* **2021**, *18*, 239–263. [\[CrossRef\]](#)
6. Chen, Y.; Liu, Z.; Han, S.; Han, J.; Jiang, D. Poly(propylene carbonate) polyurethane self-polishing coating for marine antifouling application. *J. Appl. Polym. Sci.* **2016**, *133*, 43667. [\[CrossRef\]](#)
7. Zhao, X.; Hao, H.; Duan, Y. A Self-Polishing Polyacrylate-g-polysiloxane Paint for Marine Antifouling Application. *Polym. Sci. Ser. A* **2020**, *62*, 184–195. [\[CrossRef\]](#)
8. Huggett, R.J.; Unger, M.A.; Seligman, P.F.; Valkirs, A.O. ES&T Series: The marine biocide tributyltin. Assessing and managing the environmental risks. *Environ. Sci. Technol.* **1992**, *26*, 232–237.
9. Alzieu, C. Environmental impact of TBT: The French experience. *Sci. Total Environ.* **2000**, *258*, 99–102. [\[CrossRef\]](#)
10. Feng, S.; Huang, Y.; Wang, Q.; Gao, Y.; Qing, F.-L. Synthesis and Antibiofouling Properties of Novel Crosslinkable Terpolymers Containing Semifluoroalkyl Substituted Aromatic Side Chains. *Polym. Eng. Sci.* **2010**, *50*, 944–951. [\[CrossRef\]](#)
11. Ai, X.; Xie, Q.; Ma, C.; Zhang, G. Fouling Release Coating Consisting of Hyperbranched Poly(ϵ -caprolactone)/Siloxane Elastomer. *ACS Appl. Polym. Mater.* **2020**, *2*, 1429–1437. [\[CrossRef\]](#)
12. Zhang, H.; Liang, T.; Liu, Y.; Misra, R.D.K.; Zhao, Y. Low-surface-free-energy GO/FSiAC coating with self-healing function for anticorrosion and antifouling applications. *Surf. Coat. Technol.* **2021**, *425*, 127690. [\[CrossRef\]](#)
13. Cho, J.H.; Shanmuganathan, K.; Ellison, C.J. Bioinspired catecholic copolymers for antifouling surface coatings. *ACS Appl. Mater. Interfaces* **2013**, *5*, 3794–3802. [\[CrossRef\]](#) [\[PubMed\]](#)
14. Ferreira, O.; Rijo, P.; Gomes, J.; Santos, R.; Monteiro, S.; Vilas-Boas, C.; Correia-da-Silva, M.; Almada, S.; Alves, L.; Bordado, J.; et al. Biofouling Inhibition with Grafted Econe Biocide: Toward a Nonreleasing Eco-Friendly Multiresistant Antifouling Coating. *ACS Sustain. Chem. Eng.* **2019**, *8*, 12–17. [\[CrossRef\]](#)
15. Guo, H.; Chen, P.; Tian, S.; Ma, Y.; Li, Q.; Wen, C.; Yang, J.; Zhang, L. Amphiphilic Marine Antifouling Coatings Based on a Hydrophilic Polyvinylpyrrolidone and Hydrophobic Fluorine-Silicon-Containing Block Copolymer. *Langmuir* **2020**, *36*, 14573–14581. [\[CrossRef\]](#) [\[PubMed\]](#)
16. Patterson, A.L.; Wenning, B.; Rizis, G.; Calabrese, D.R.; Finlay, J.A.; Franco, S.C.; Zuckermann, R.N.; Clare, A.S.; Kramer, E.J.; Ober, C.K.; et al. Role of Backbone Chemistry and Monomer Sequence in Amphiphilic Oligopeptide- and Oligopeptoid-Functionalized PDMS- and PEO-Based Block Copolymers for Marine Antifouling and Fouling Release Coatings. *Macromolecules* **2017**, *50*, 2656–2667. [\[CrossRef\]](#)

17. Ai, X.; Mei, L.; Ma, C.; Zhang, G. Degradable hyperbranched polymer with fouling resistance for antifouling coatings. *Prog. Org. Coat.* **2021**, *153*, 106141. [[CrossRef](#)]
18. Yao, J.; Dai, Z.; Yi, J.; Yu, H.; Wu, B.; Dai, L. Degradable polyurethane based on triblock polyols composed of polypropylene glycol and ϵ -caprolactone for marine antifouling applications. *J. Coat. Technol. Res.* **2020**, *17*, 865–874. [[CrossRef](#)]
19. Chen, X.; Zhang, G.; Zhang, Q.; Zhan, X.; Chen, F. Preparation and Performance of Amphiphilic Polyurethane Copolymers with Capsaicin-Mimic and PEG Moieties for Protein Resistance and Antibacteria. *Ind. Eng. Chem. Res.* **2015**, *54*, 3813–3820. [[CrossRef](#)]
20. Wang, F.; Zhang, H.; Yu, B.; Wang, S.; Shen, Y.; Cong, H. Review of the research on anti-protein fouling coatings materials. *Prog. Org. Coat.* **2020**, *147*, 105860. [[CrossRef](#)]
21. Tian, L.; Yin, Y.; Jin, H.; Bing, W.; Jin, E.; Zhao, J.; Ren, L. Novel marine antifouling coatings inspired by corals. *Mater. Today Chem.* **2020**, *17*, 100294. [[CrossRef](#)]
22. Xu, L.P.; Peng, J.; Liu, Y.; Wen, Y.; Zhang, X.; Jiang, L.; Wang, S. Nacre-inspired design of mechanical stable coating with underwater superoleophobicity. *ACS Nano* **2013**, *7*, 5077–5083. [[CrossRef](#)] [[PubMed](#)]
23. Kim, K.S.; Gunari, N.; MacNeil, D.; Finlay, J.; Callow, M.; Callow, J.; Walker, G.C. Aqueous-Based Fabrication of Low-VOC Nanostructured Block Copolymer Films as Potential Marine Antifouling Coatings. *ACS Appl. Mater. Interfaces* **2016**, *8*, 20342–20351. [[CrossRef](#)] [[PubMed](#)]
24. Liu, Z.; Tian, S.; Li, Q.; Wang, J.; Pu, J.; Wang, G.; Zhao, W.; Feng, F.; Qin, J.; Ren, L. Integrated Dual-Functional ORMOSIL Coatings with AgNPs@rGO Nanocomposite for Corrosion Resistance and Antifouling Applications. *ACS Sustain. Chem. Eng.* **2020**, *8*, 6786–6797. [[CrossRef](#)]
25. Wanka, R.; Koc, J.; Clarke, J.; Hunsucker, K.Z.; Swain, G.W.; Aldred, N.; Finlay, J.A.; Clare, A.S.; Rosenhahn, A. Sol-Gel-Based Hybrid Materials as Antifouling and Fouling-Release Coatings for Marine Applications. *ACS Appl. Mater. Interfaces* **2020**, *12*, 53286–53296. [[CrossRef](#)]
26. Su, X.; Hao, D.; Xu, X.; Guo, X.; Li, Z.; Jiang, L. Hydrophilic/Hydrophobic Heterogeneity Anti-Biofouling Hydrogels with Well-Regulated Rehydration. *ACS Appl. Mater. Interfaces* **2020**, *12*, 25316–25323. [[CrossRef](#)]
27. Wu, G.; Li, C.-C.; Jiang, X.-H.; Yu, L.-M. Highly efficient antifouling property based on self-generating hydrogel layer of polyacrylamide coatings. *J. Appl. Polym. Sci.* **2016**, *133*, 44111. [[CrossRef](#)]
28. Tian, J.; Xu, K.; Hu, J.; Zhang, S.; Cao, G.; Shao, G. Durable self-polishing antifouling Cu-Ti coating by a micron-scale Cu/Ti laminated microstructure design. *J. Mater. Sci. Technol.* **2021**, *79*, 62–74. [[CrossRef](#)]
29. Kim, B.-W.; Kang, T.-W.; Park, H.; Lee, I.; Chun, H.; Jo, N.-J. Self-polishing behavior of zinc-based copolymer with different monomer composition. *Macromol. Res.* **2014**, *22*, 978–982. [[CrossRef](#)]
30. Chen, R.; Li, Y.; Yan, M.; Sun, X.; Han, H.; Li, J.; Wang, J.; Liu, L.; Takahashi, K. Synthesis of Hybrid Zinc/Silyl Acrylate Copolymers and their Surface Properties in the Microfouling Stage. *RSC Adv.* **2016**, *6*, 13858–13866. [[CrossRef](#)]
31. Dai, Z.; Cao, M.; Li, S.; Yao, J.; Wu, B.; Wang, Y.; Wang, H.; Dong, J.; Yi, J. A novel marine antifouling coating based on a self-polishing zinc-polyurethane copolymer. *J. Coat. Technol. Res.* **2021**, *18*, 1333–1343. [[CrossRef](#)]
32. Feng, K.; Ni, C.; Yu, L.; Zhou, W.; Li, X. Synthesis and evaluation of acrylate resins suspending indole derivative structure in the side chain for marine antifouling. *Colloids Surf. B* **2019**, *184*, 110518. [[CrossRef](#)] [[PubMed](#)]
33. Yoshimura, E.; Nogata, Y.; Sakaguchi, I. Simple methods for mass culture of barnacle larvae. *Sess. Org.* **2006**, *23*, 39–42. [[CrossRef](#)]
34. Özlem, S.; Hacaloglu, J. Thermal degradation of poly(n-butyl methacrylate), poly(n-butyl acrylate) and poly(t-butyl acrylate). *J. Anal. Appl. Pyrolysis* **2013**, *104*, 161–169. [[CrossRef](#)]
35. Zhang, X.; Ding, Y.; Zhang, G. Preparation and Characterization of Solvent Based Polyacrylate Pressure Sensitive Adhesives. *Chem. Adhes.* **2011**, *10*, 30–34.
36. Lejars, M.; Margaillan, A.; Bressy, C. Fouling Release Coatings: A Nontoxic Alternative to Biocidal Antifouling Coatings. *Chem. Rev.* **2012**, *112*, 4347–4390. [[CrossRef](#)] [[PubMed](#)]
37. Dong, M.; Liu, Z.; Gao, Y.; Wang, X.; Chen, J.; Yang, J. Synergistic effect of copolymeric resin grafted 1,2-benzisothiazol-3(2H)-one and heterocyclic groups as a marine antifouling coating. *RSC Adv.* **2021**, *11*, 18787–18796. [[CrossRef](#)]
38. Gao, L.; Zhang, H.; Fan, J.; Zhou, B.; Wang, Y.; Wu, C. Inhibitional effect of 3 chemical substances on *chlorella*. *Chin. J. Environ. Eng.* **2013**, *7*, 4869–4873.
39. Wang, X.-X.; Zhang, T.-Y.; Dao, G.-H.; Hu, H.-Y. Interaction between 1,2-benzisothiazol-3(2H)-one and microalgae: Growth inhibition and detoxification mechanism. *Aquat. Toxicol.* **2018**, *205*, 66–75. [[CrossRef](#)]
40. Morley, J.O.; Oliver, A.J.; Charlton, M.H. Theoretical studies on the biocidal activity of 5-chloro-3-isothiazolone. *J. Mol. Struct.* **1998**, *429*, 103–110. [[CrossRef](#)]
41. Makino, K.; Oshima, K.; Kurokawa, K.; Yokoyama, K.; Uda, T.; Tagomori, K.; Iijima, Y.; Najima, M.; Nakano, M.; Yamashita, A.; et al. Genome sequence of *Vibrio parahaemolyticus*: A pathogenic mechanism distinct from that of *V. cholerae*. *Lancet* **2003**, *361*, 743–749. [[CrossRef](#)]
42. Sung, C.-G.; Choi, M.-S.; Chun, H.H.; Lee, I.; Park, H.S.; Lee, C.-H. Effect of Heavy Metals on Larvae of Barnacle, *Balanus albicostatus*. *Environ. Anal. Health Toxicol.* **2008**, *23*, 337–343.